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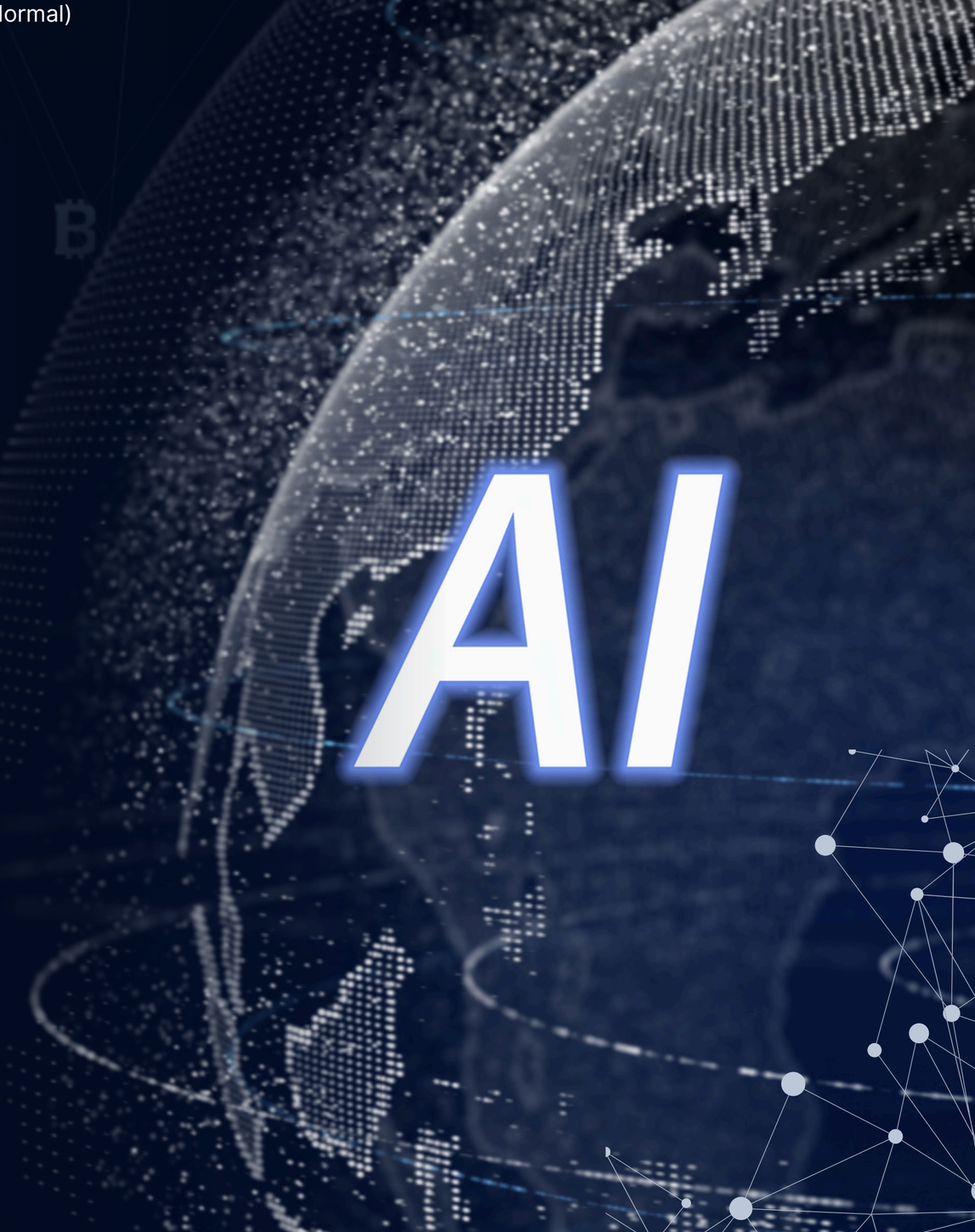
# DOAA

# CA2 PART B

## MLOPS FOR DL WEB APPLICATION




Presented By

**Goh Kun Ming**

*https://orcid.org/0009-0008-7666-781X*

Admin Number

**P2415691**

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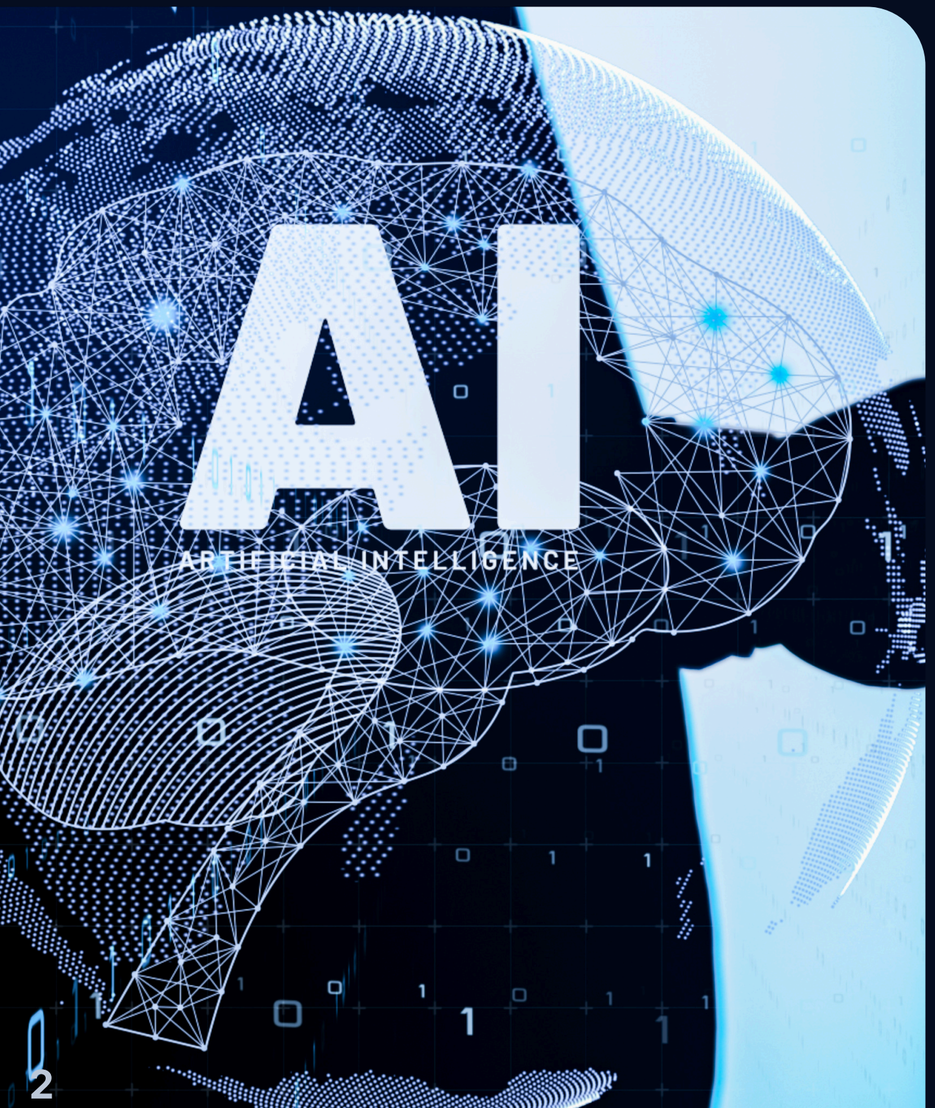
# BUSINESS VALUE




- Faster vegetable identification for consumers and staff (image classifier + live detection).
- Better buying decisions through confidence-scored predictions, not guesswork.
- Real-time support at point-of-sale or market visits with camera-based live detection.
- Price transparency via Price Evaluator and searchable/filterable price views.
- Smarter shopping and budgeting by comparing produce options quickly.
- RAG assistant gives contextual guidance (produce info, usage, selection support) grounded in your app data.
- Reduced manual effort for identifying characteristics in produce.
- Improved consistency and accuracy versus human-only visual checks.
- Stronger trust and compliance posture (auth, captcha/OTP, security gates in CI/CD).
- Deployment-ready architecture (containerized services + automated pipeline + Render deploy hooks).

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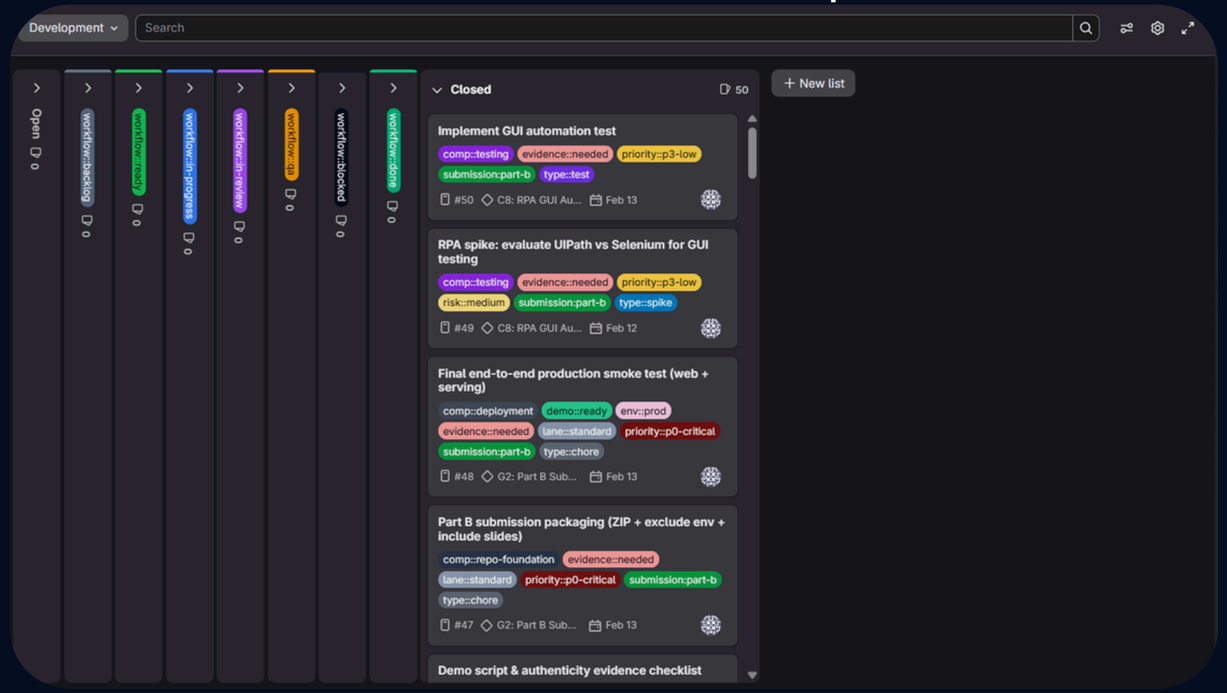
# DEVOPS PROCESS

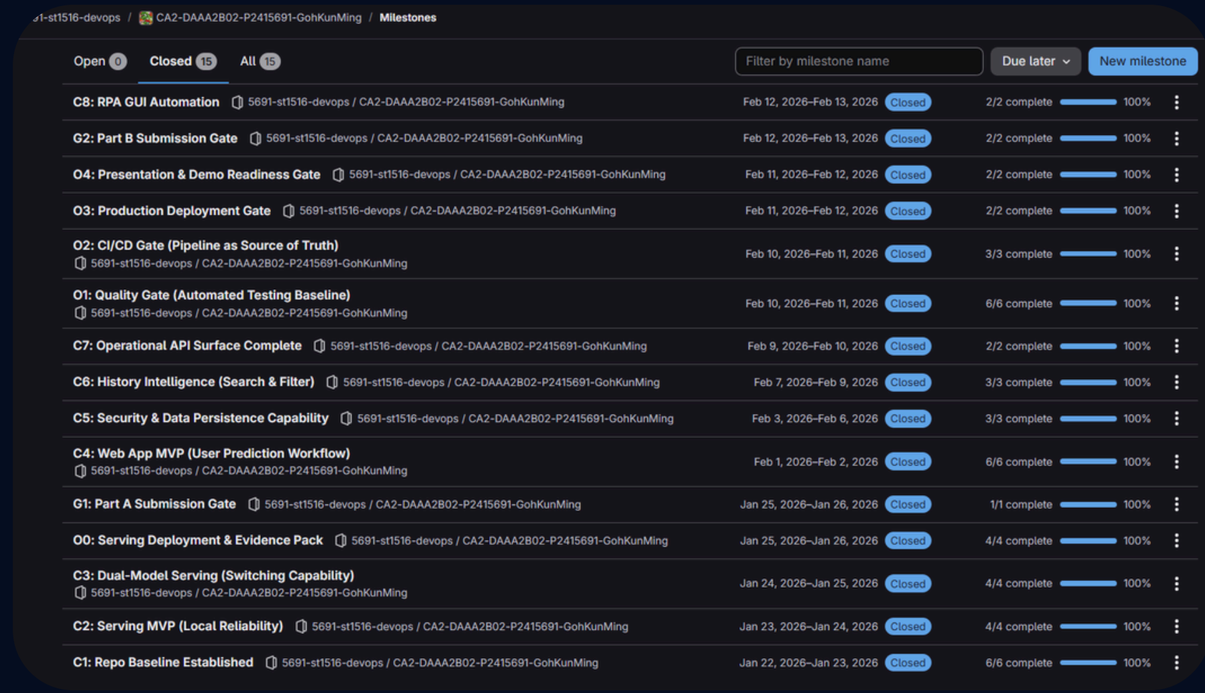
## GitLab Information

178 Commits
22 Branches
0 Tags
52 MiB Project Storage

## Scrum Board Setup




## Milestones




C: Content
O: Operation
G: Gate

https://gitlab.com/5691-st1516-devops/ca2-daaa2b02-p2415691-gohkunming

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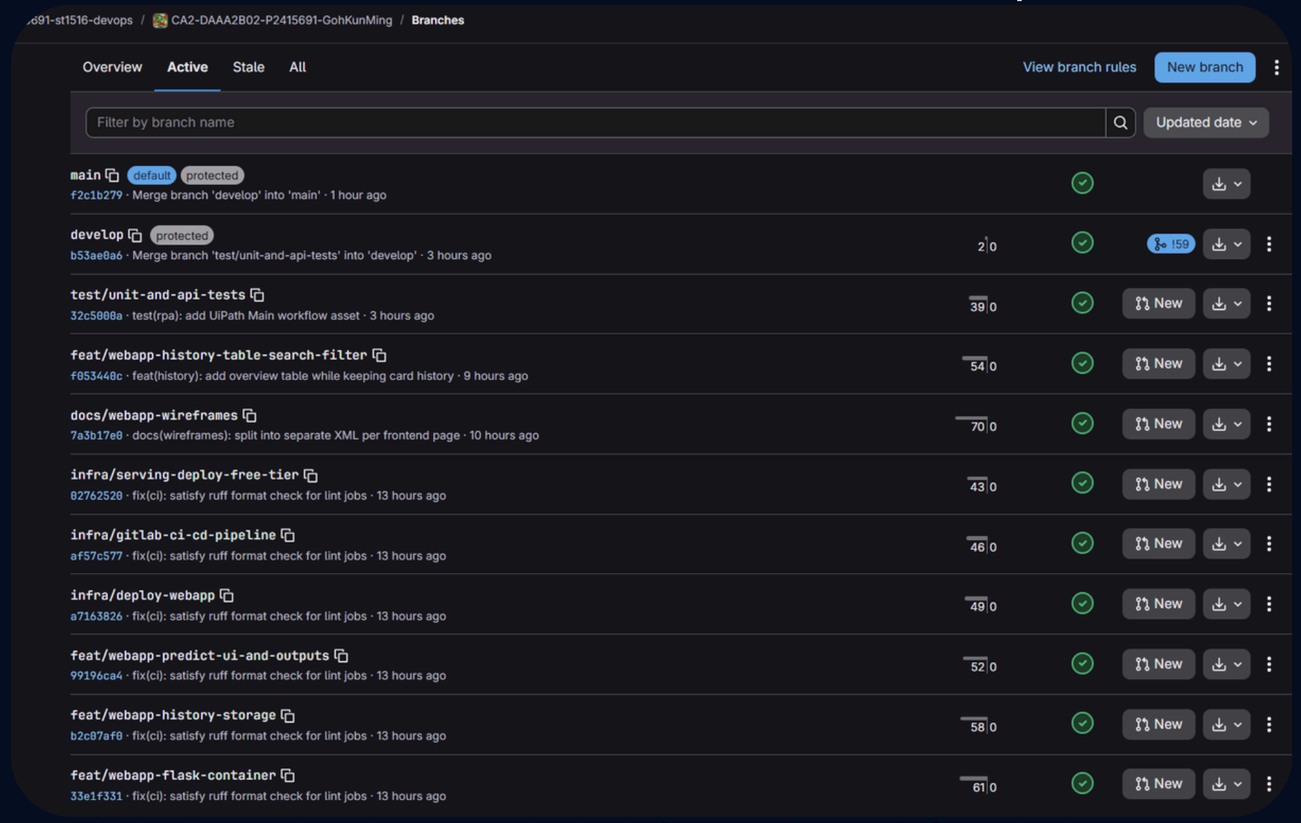
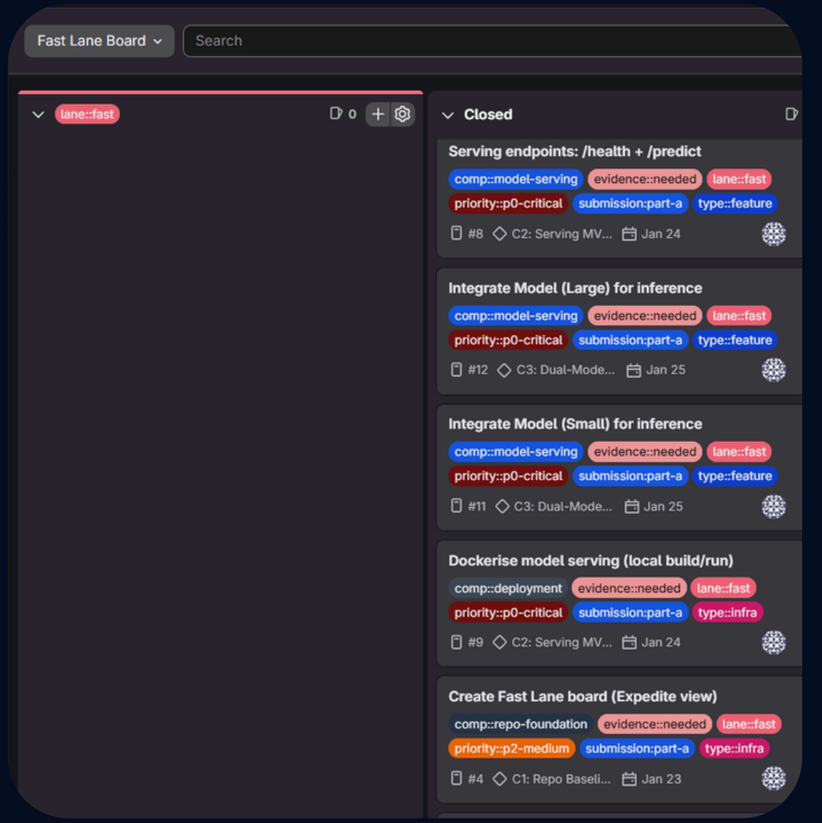
# DEVOPS PROCESS

## Branches (20 excl. main / develop)




All development branch merge to develop before merging to main for each release gate (Submission A & Submission B)

## Fast Lane




Used primarily for Part A due to urgency

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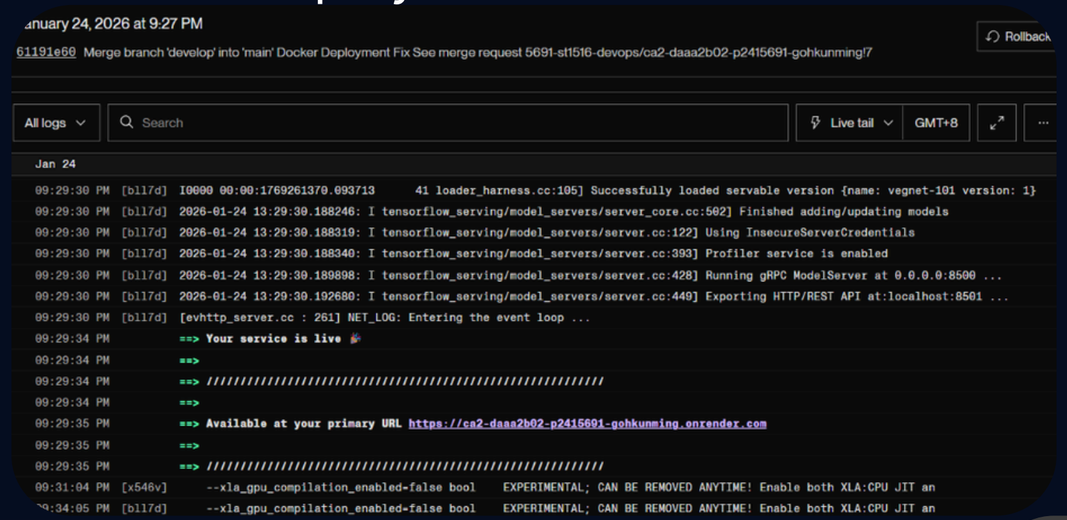
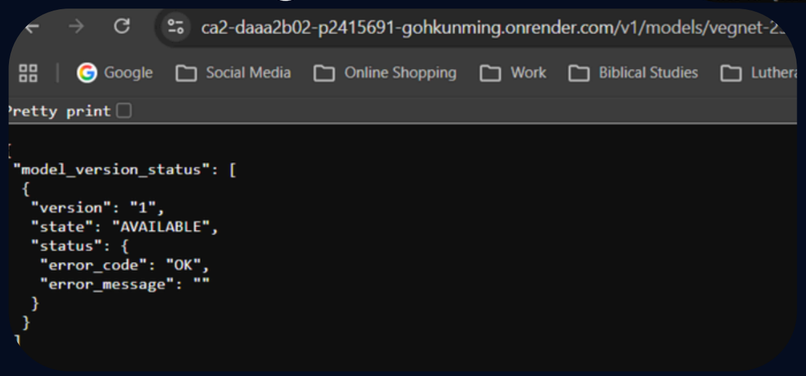
# MODEL SERVICE

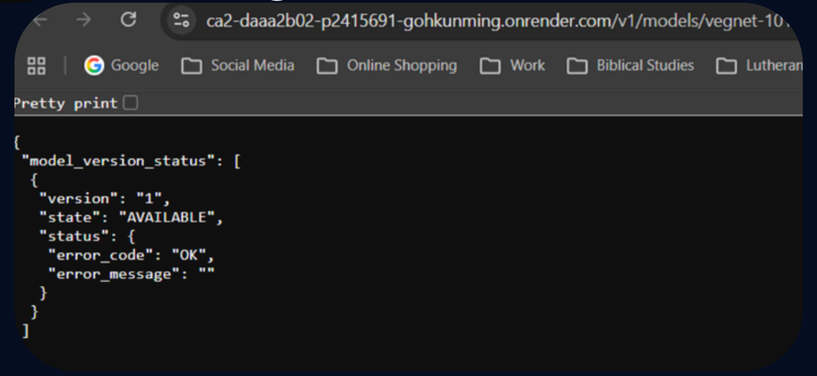
## Deployment on Render




### Vegnet (23x23)




### Vegnet (101x101)




https://ca2-daaa2b02-p2415691-gohkunming.onrender.com/v1/models/vegnet-23
https://ca2-daaa2b02-p2415691-gohkunming.onrender.com/v1/models/vegnet-101

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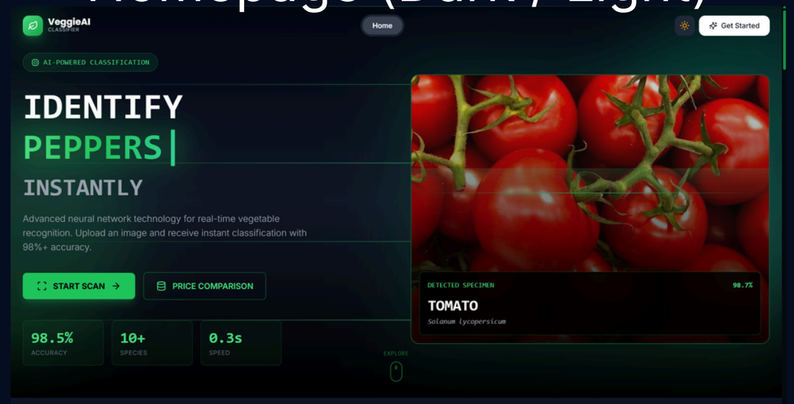
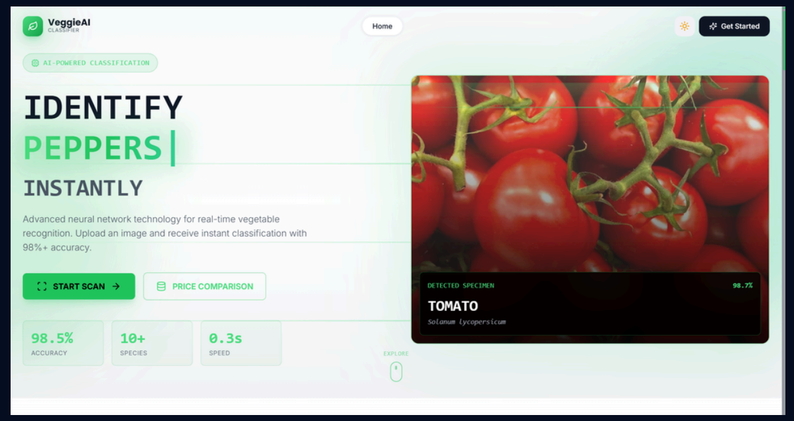
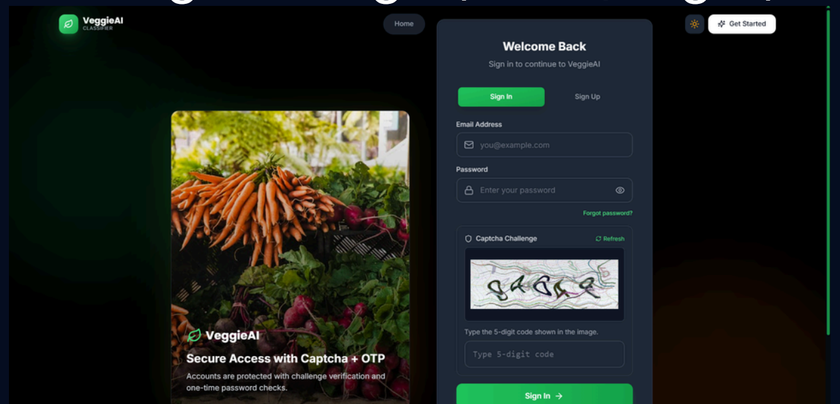
# WEB APPLICATION DEVELOPMENT

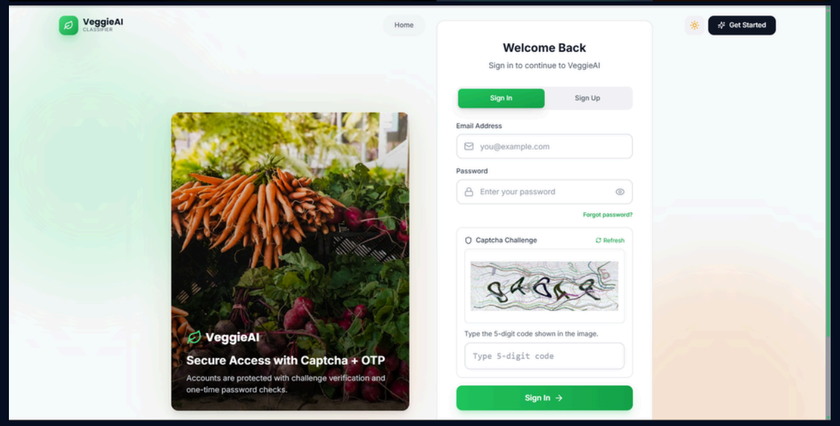
## Homepage (Dark / Light)



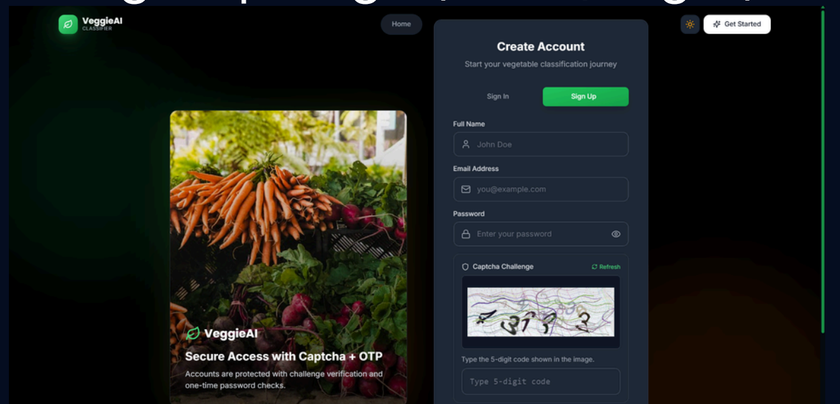



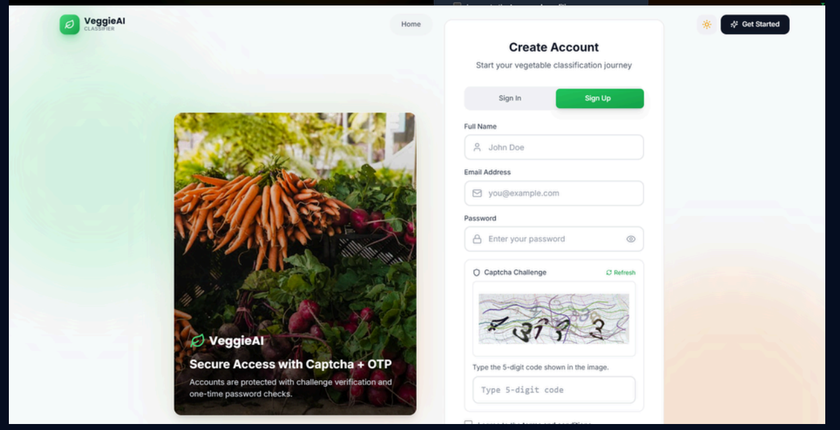

## Sign In Page (Dark / Light)




## Sign Up Page (Dark / Light)




*https://ca2-daaa2b02-p2415691-gohkunming-1.onrender.com*

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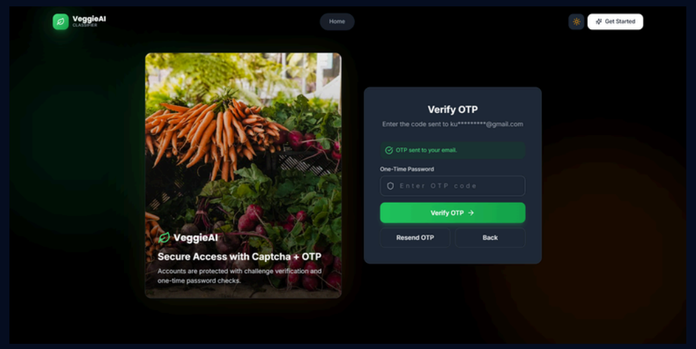
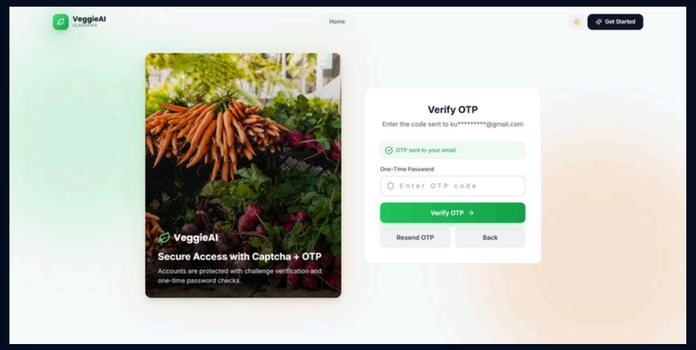
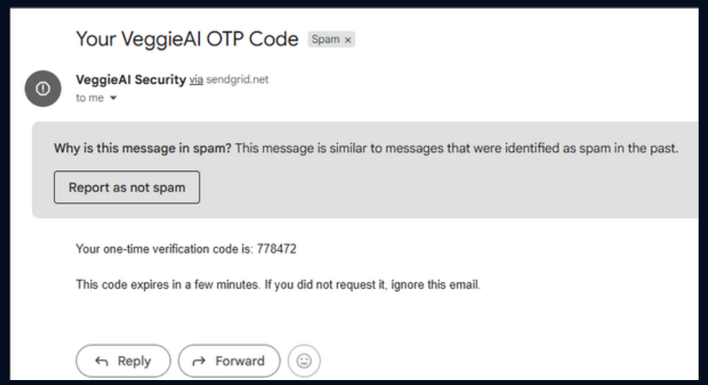
# WEB APPLICATION DEVELOPMENT

## OTP (Dark / Light)

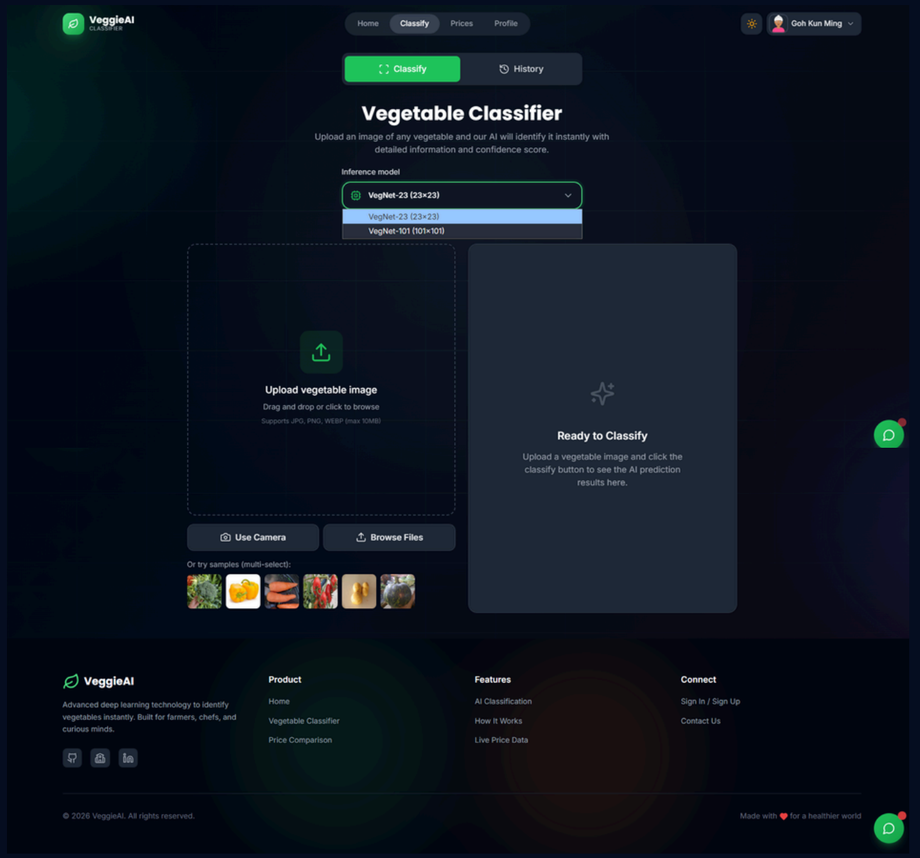



## OTP Email (SMTP)




## Prediction Page (Dark / Light)

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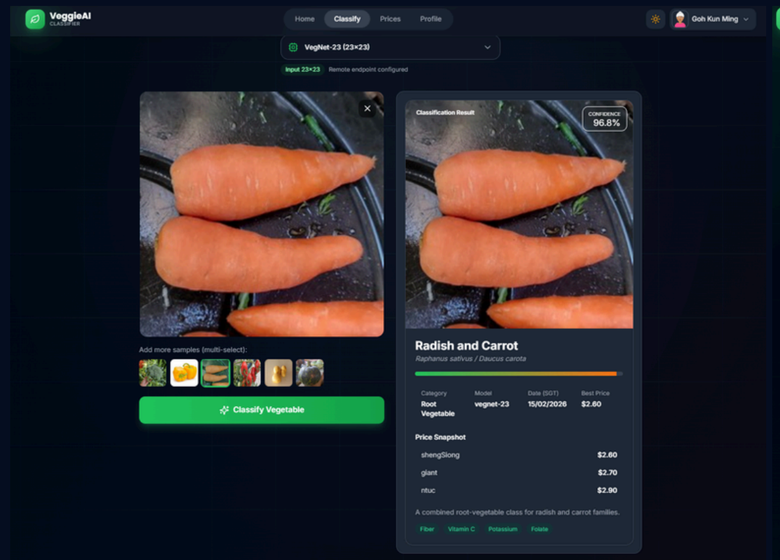
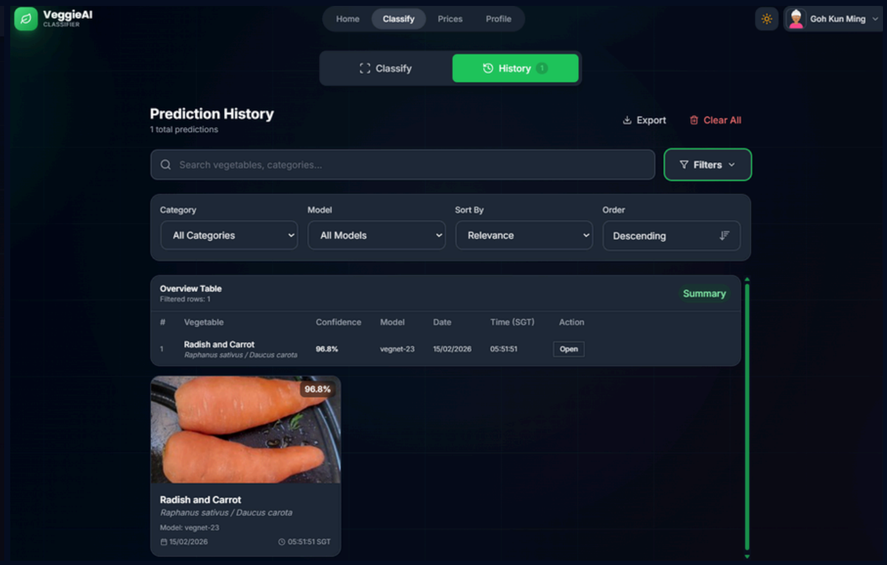
# WEB APPLICATION DEVELOPMENT

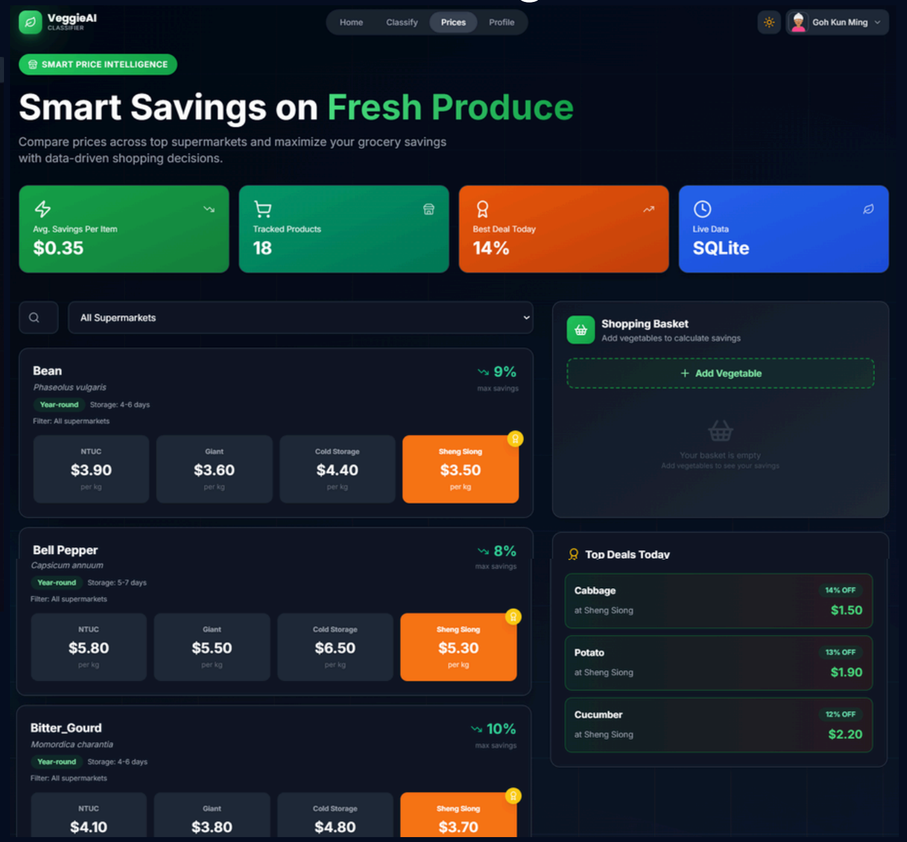
## Prediction




## History & Search / Filter




## Price Intelligence

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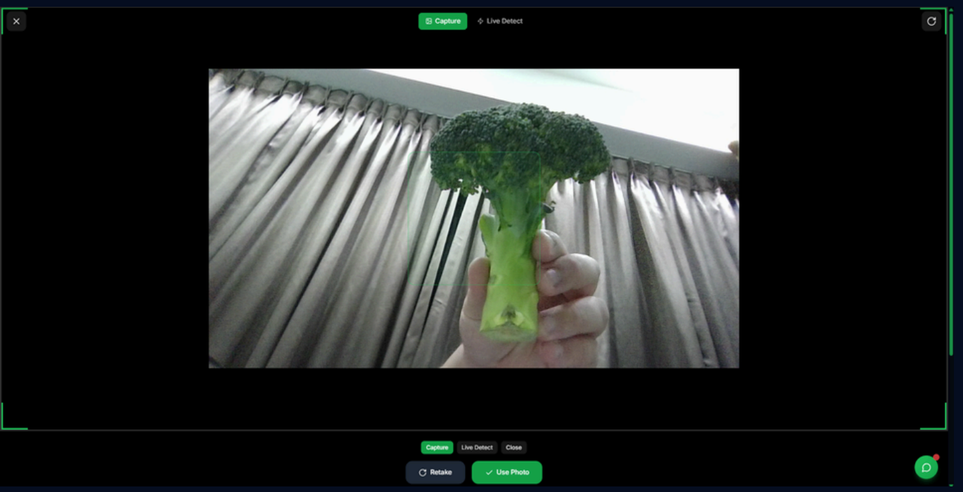
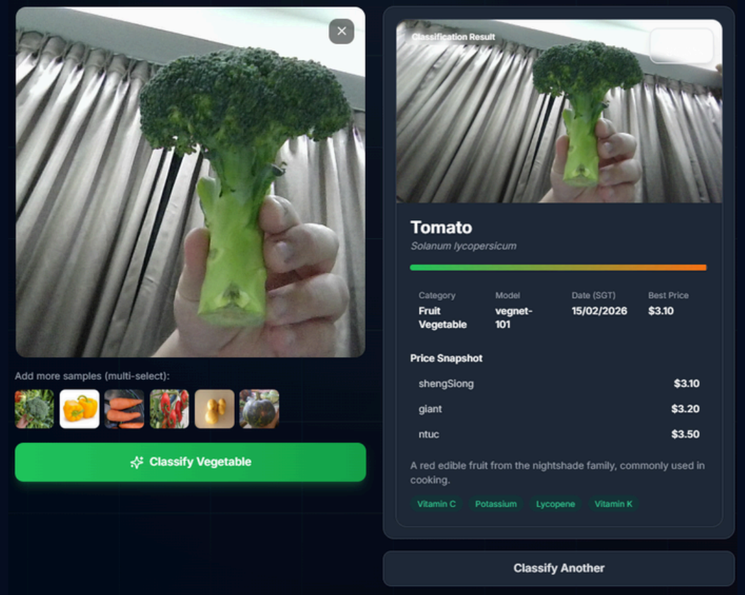
# WEB APPLICATION DEVELOPMENT

## Camera Capture




## Prediction using Camera Captured Image




Poor Accuracy due to Image Noise / Poor Model

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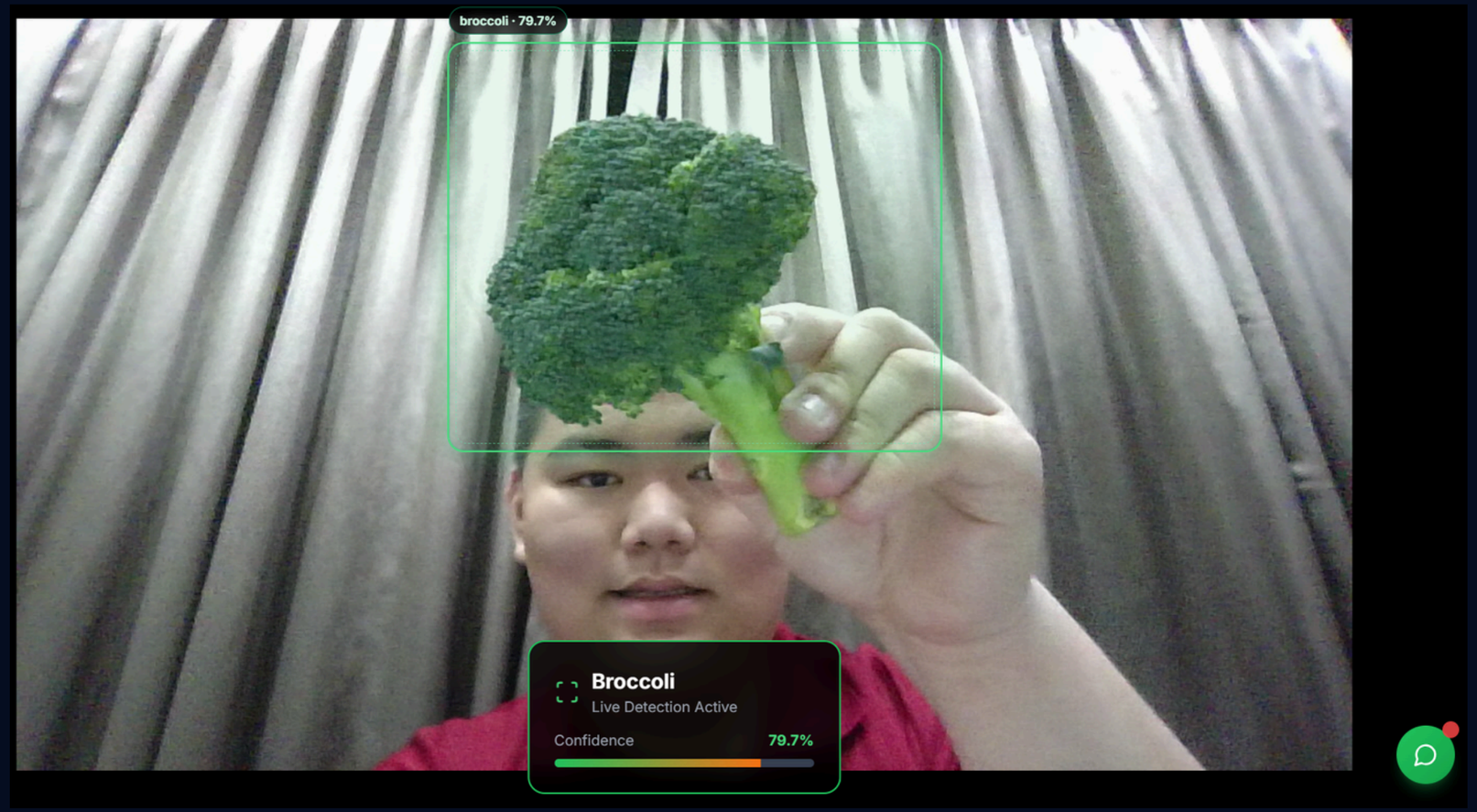
# WEB APPLICATION DEVELOPMENT

## Live Detection




https://ca2-daaa2b02-p2415691-gohkunming-1.onrender.com

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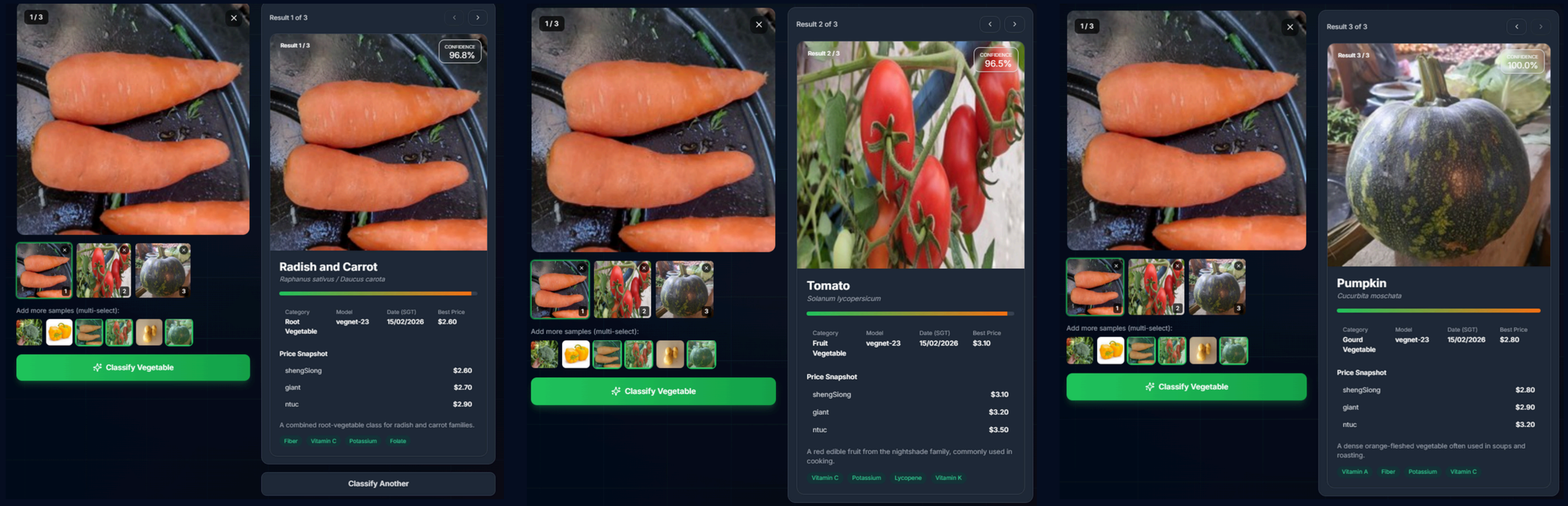
# WEB APPLICATION DEVELOPMENT

## Model Batch Prediction




https://ca2-daaa2b02-p2415691-gohkunming-1.onrender.com

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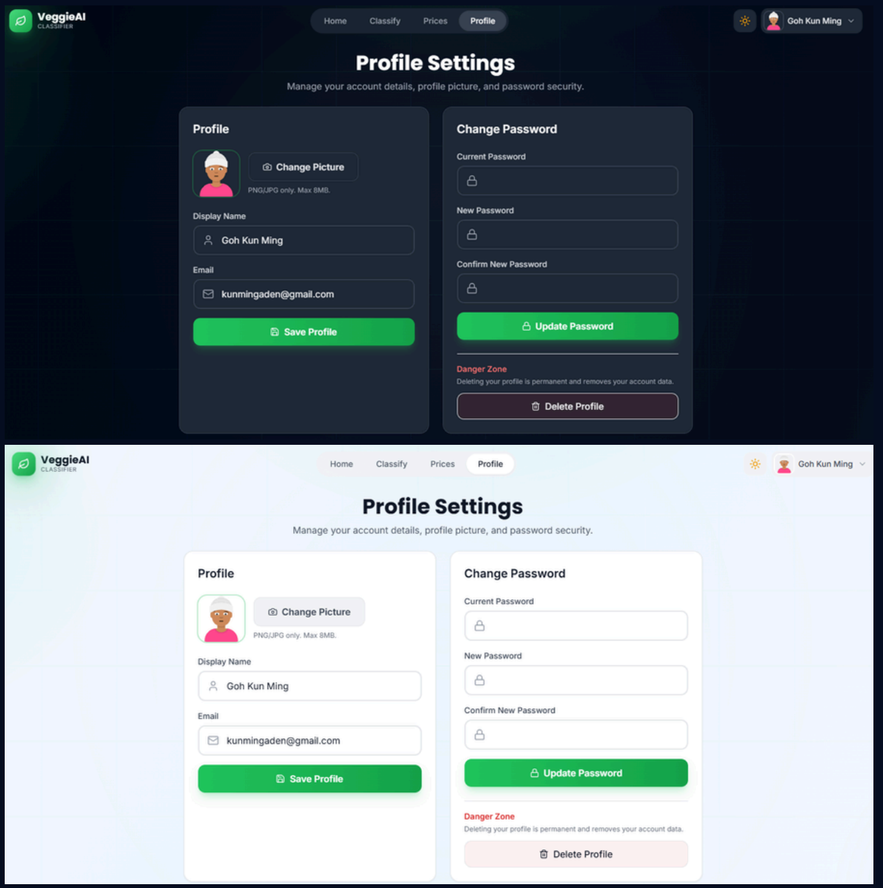
# WEB APPLICATION DEVELOPMENT

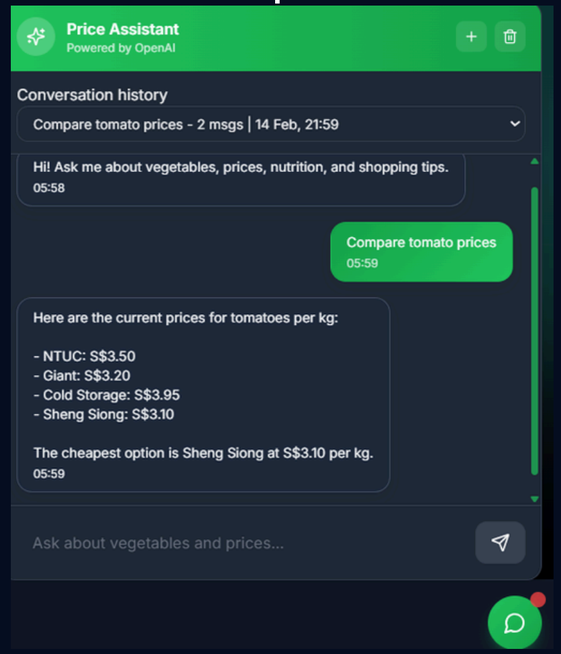
## Profile



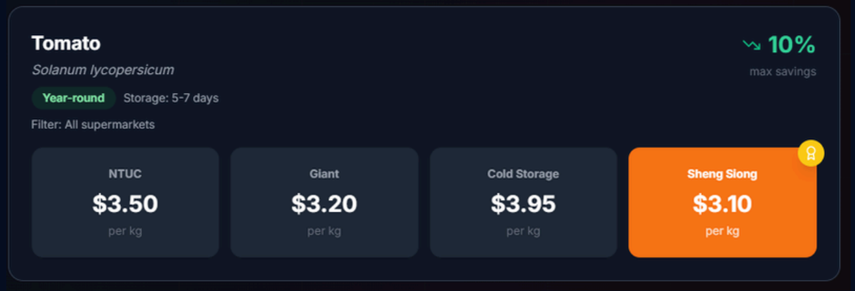




## RAG-Capabilities




## Source of Truth Reference

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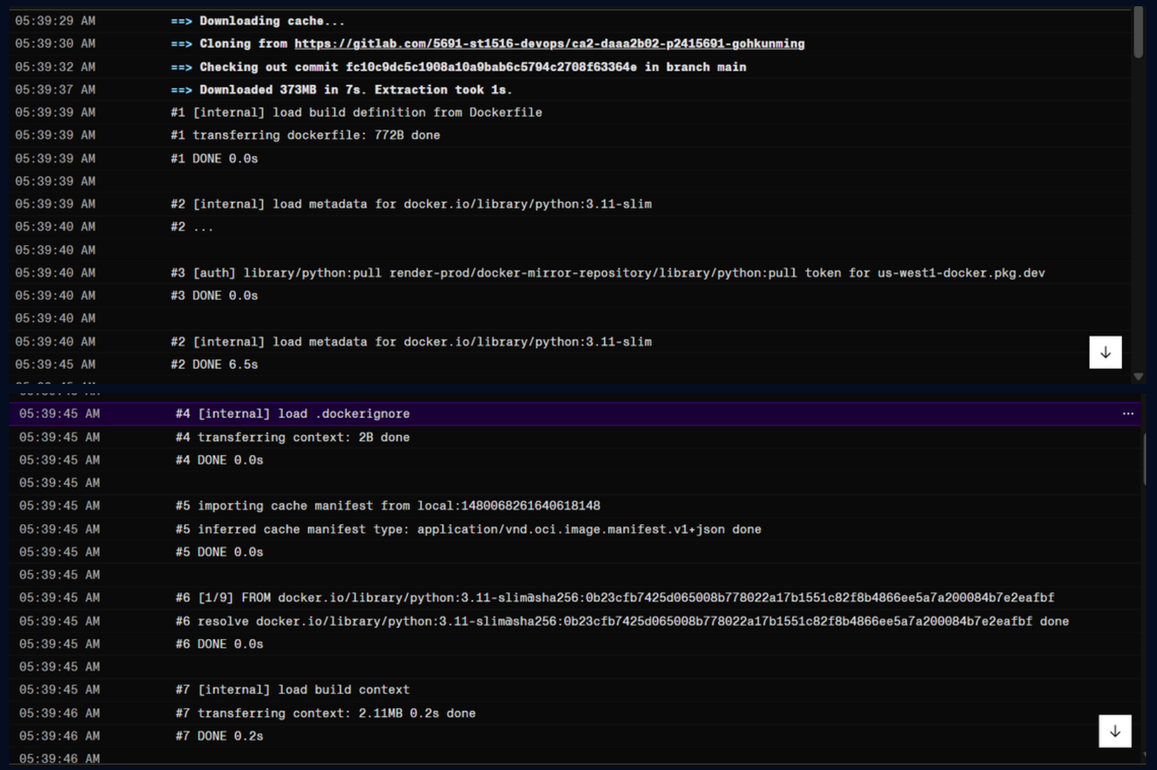
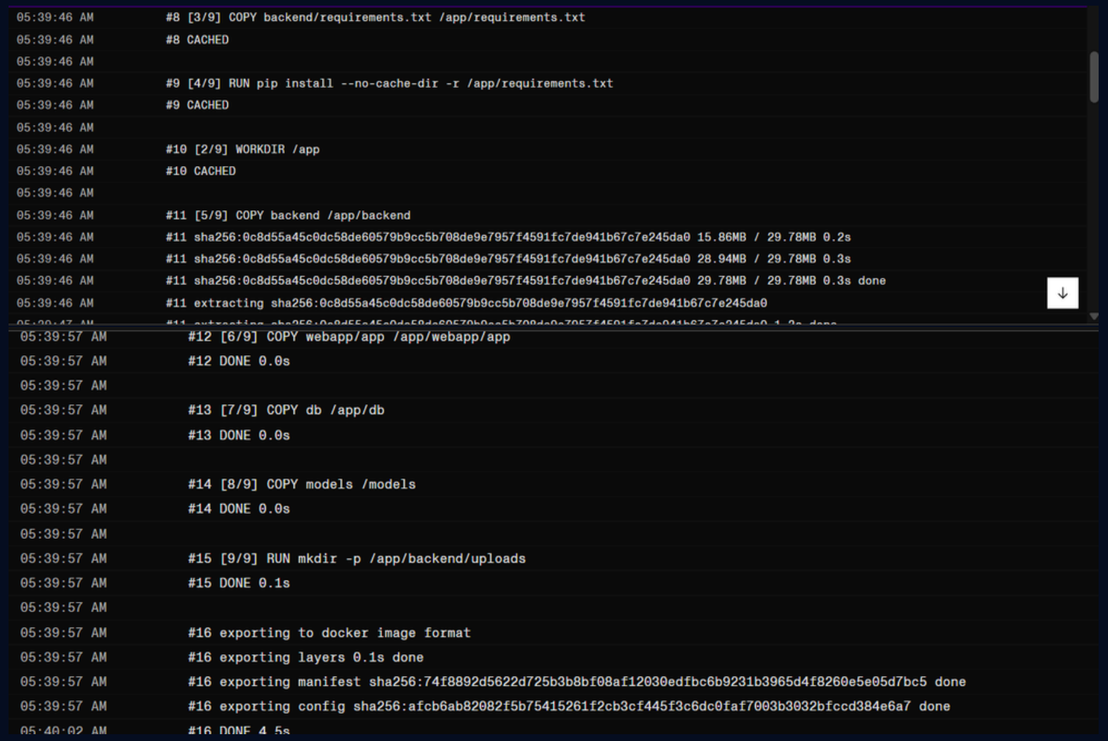
# WEB APPLICATION DEVELOPMENT

## Docker Deployment Logs

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# WEB APPLICATION DEVELOPMENT

## Docker Deployment Success Logs

```
05:40:05 AM          Upload succeeded
05:40:10 AM          ==> Deploying...
05:40:11 AM          ==> Setting WEB_CONCURRENCY=1 by default, based on available CPUs in the instance
05:40:48 AM  [kdtjb]  [2026-02-14 21:40:48 +0000] [7] [INFO] Starting gunicorn 22.0.0
05:40:48 AM  [kdtjb]  [2026-02-14 21:40:48 +0000] [7] [INFO] Listening at: http://0.0.0.0:10000 (7)
05:40:48 AM  [kdtjb]  [2026-02-14 21:40:48 +0000] [7] [INFO] Using worker: gthread
05:40:48 AM  [kdtjb]  [2026-02-14 21:40:48 +0000] [8] [INFO] Booting worker with pid: 8
05:40:48 AM  [kdtjb]  [2026-02-14 21:40:48 +0000] [9] [INFO] Booting worker with pid: 9
05:40:51 AM  [kdtjb]  127.0.0.1 - - [14/Feb/2026:21:40:51 +0000] "HEAD / HTTP/1.1" 308 0 "-" "Go-http-client/1.1"
05:40:59 AM          ==> Your service is live 🎉
05:40:59 AM          ==>
05:40:59 AM          ==> ///////////////////////////////////////////////////////////
05:40:59 AM          ==>
05:40:59 AM          ==> Available at your primary URL https://ca2-daaa2b02-p2415691-gohkunming-1.onrender.com
05:40:59 AM          ==>
05:40:59 AM          ==> ///////////////////////////////////////////////////////////
```

## SQLite Database Structure

| Name | Type | Schema |
|---|---|---|
| Tables (9) | | |
| captcha_challenges | | CREATE TABLE captcha_challenges ( id TEXT PRIMARY KEY, answer_hash TEXT NOT NULL, expires_at INTEGER NOT NULL, created_at INTEGER NOT NULL ) |
| chat_conversations | | CREATE TABLE chat_conversations ( id INTEGER PRIMARY KEY AUTOINCREMENT, user_id INTEGER NOT NULL, title TEXT NOT NULL DEFAULT 'New chat', created_at TEXT NOT NULL DEFAULT |
| chat_messages | | CREATE TABLE chat_messages ( id INTEGER PRIMARY KEY AUTOINCREMENT, user_id INTEGER NOT NULL, conversation_id INTEGER NOT NULL, role TEXT NOT NULL CHECK(role IN ('user', 'as |
| otp_sessions | | CREATE TABLE otp_sessions ( session_id TEXT PRIMARY KEY, user_id INTEGER NOT NULL, purpose TEXT NOT NULL, email TEXT NOT NULL, otp_hash TEXT NOT NULL, attempts INTEGER NOT |
| predictions | | CREATE TABLE predictions ( id INTEGER PRIMARY KEY AUTOINCREMENT, user_id INTEGER NOT NULL, created_at TEXT NOT NULL DEFAULT (datetime('now')), model_name TEXT NOT NULL, i |
| sqlite_sequence | | CREATE TABLE sqlite_sequence(name,seq) |
| users | | CREATE TABLE users ( id INTEGER PRIMARY KEY AUTOINCREMENT, username TEXT NOT NULL UNIQUE, password_hash TEXT NOT NULL, email TEXT, display_name TEXT, avatar TEXT, role TI |
| vegetables | | CREATE TABLE vegetables ( key TEXT PRIMARY KEY, name TEXT NOT NULL, scientific_name TEXT NOT NULL, category TEXT NOT NULL, description TEXT NOT NULL, nutrients_json TEXT NOT |
| veggie_predictions | | CREATE TABLE veggie_predictions ( id TEXT PRIMARY KEY, user_id INTEGER NOT NULL, image_ref TEXT NOT NULL, image_sha256 TEXT NOT NULL, model_name TEXT NOT NULL DEFAULT 'v |
| Indices (15) | | |
| idx_captcha_challenges_expires | | CREATE INDEX idx_captcha_challenges_expires ON captcha_challenges(expires_at) |
| idx_chat_conversations_user_last_mes... | | CREATE INDEX idx_chat_conversations_user_last_message ON chat_conversations(user_id, last_message_at DESC) |
| idx_chat_messages_user_conversation... | | CREATE INDEX idx_chat_messages_user_conversation_created ON chat_messages(user_id, conversation_id, created_at) |
| idx_chat_messages_user_created | | CREATE INDEX idx_chat_messages_user_created ON chat_messages(user_id, created_at) |
| idx_otp_sessions_expires | | CREATE INDEX idx_otp_sessions_expires ON otp_sessions(expires_at) |
| idx_otp_sessions_user_id | | CREATE INDEX idx_otp_sessions_user_id ON otp_sessions(user_id) |
| idx_predictions_created_at | | CREATE INDEX idx_predictions_created_at ON predictions(created_at) |
| idx_predictions_model_name | | CREATE INDEX idx_predictions_model_name ON predictions(model_name) |
| idx_predictions_sha256 | | CREATE INDEX idx_predictions_sha256 ON predictions(sha256) |
| idx_predictions_top1_label | | CREATE INDEX idx_predictions_top1_label ON predictions(top1_label) |
| idx_predictions_user_id | | CREATE INDEX idx_predictions_user_id ON predictions(user_id) |
| idx_users_email_unique | | CREATE UNIQUE INDEX idx_users_email_unique ON users(email) WHERE email IS NOT NULL |
| idx_veggie_predictions_created_at | | CREATE INDEX idx_veggie_predictions_created_at ON veggie_predictions(created_at) |
| idx_veggie_predictions_model_name | | CREATE INDEX idx_veggie_predictions_model_name ON veggie_predictions(model_name) |
| idx_veggie_predictions_user_id | | CREATE INDEX idx_veggie_predictions_user_id ON veggie_predictions(user_id) |
| Views (0) | | |
| Triggers (0) | | |

## Password Hashing + Salting (One-Way Cryptography)

| id | username | password_hash | email | display_name | avatar | role | created_at |
|---|---|---|---|---|---|---|---|
| Fil... | Filter | Filter | Filter | Filter | Filter | Filter | Filter |
| 1 | admin | $2b$12$VsJCa5RWeyS2yN0pQ60hUO/… | *NULL* | *NULL* | *NULL* | admin | 2026-02-11 13:05:03 |
| 3 | kunmingaden | $2b$12$BMkAkxE/yubvQGjU.… | kunmingaden@gmail.com | Goh Kun Ming | https://api.dicebear.com/7.x/… | user | 2026-02-12 16:43:52 |

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# AUTOMATIC TESTING

## Overall Test

```
backend\tests\test_api_contract_routes.py .....................                [  9%]
backend\tests\test_auth.py ..................................                  [ 26%]
backend\tests\test_chat.py ................                                    [ 33%]
backend\tests\test_e2e_flow.py .                                               [ 33%]
backend\tests\test_health.py .                                                 [ 34%]
backend\tests\test_history.py ......                                           [ 37%]
backend\tests\test_models.py .                                                 [ 37%]
backend\tests\test_part3_unit_requirements.py ..............                   [ 44%]
backend\tests\test_predict.py ........                                         [ 48%]
backend\tests\test_security_hardening.py ...........                           [ 53%]
backend\tests\test_security_penetration.py ........................            [ 66%]
backend\tests\test_smoke_scripts.py ...                                        [ 67%]
backend\tests\test_utils_files.py ........                                     [ 71%]
backend\tests\test_veggie_api.py ........................                      [ 83%]
backend\tests\test_views_jinja.py ................                             [ 92%]
model_service\tests\test_health.py .                                           [ 92%]
model_service\tests\test_model_validation.py ...                               [ 94%]
model_service\tests\test_predict.py ............                               [100%]

================================== 206 passed in 109.82s (0:01:49) ==================================
```

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# AUTOMATIC TESTING

**Unit tests**

- Core validation/preprocessing logic (image parsing, input-size rules, probability extraction, deterministic outputs).
- Covers validity, range, consistency, expected-failure, and unexpected-failure scenarios.

**API tests (backend + model service)**

- Auth APIs: login/signup/register/logout/me, captcha/OTP, password reset, profile flows.
- Prediction APIs: /api/predict, /api/veggie/predict, /api/veggie/detect.
- History APIs: list/detail/image/delete, pagination/filter bounds, user isolation.
- Prices/catalog/model list APIs.
- Chat/RAG APIs: message, history, conversations.
- Model service APIs: /health, /predict (multipart/base64, invalid model/input, top_k validation).

**Contract tests**

- Route registration test to ensure crucial REST endpoints exist and are correctly exposed.

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# AUTOMATIC TESTING

**Security tests**

- Auth guarding and unauthorized access rejection.
- Brute-force/rate-limit and invalid credential behavior.
- CORS/security-header hardening checks.
- IDOR-style access controls (cross-user access blocked).
- Captcha/OTP misuse/replay and negative-path scenarios.
- Vulnerability scanning in CI: Bandit, pip-audit, gitleaks.

**E2E tests**

- Full secure flow: captcha -> signup/login -> OTP verify -> model predict -> history -> chat interaction.
- Includes cross-boundary behavior between auth, prediction, persistence, and chat modules.

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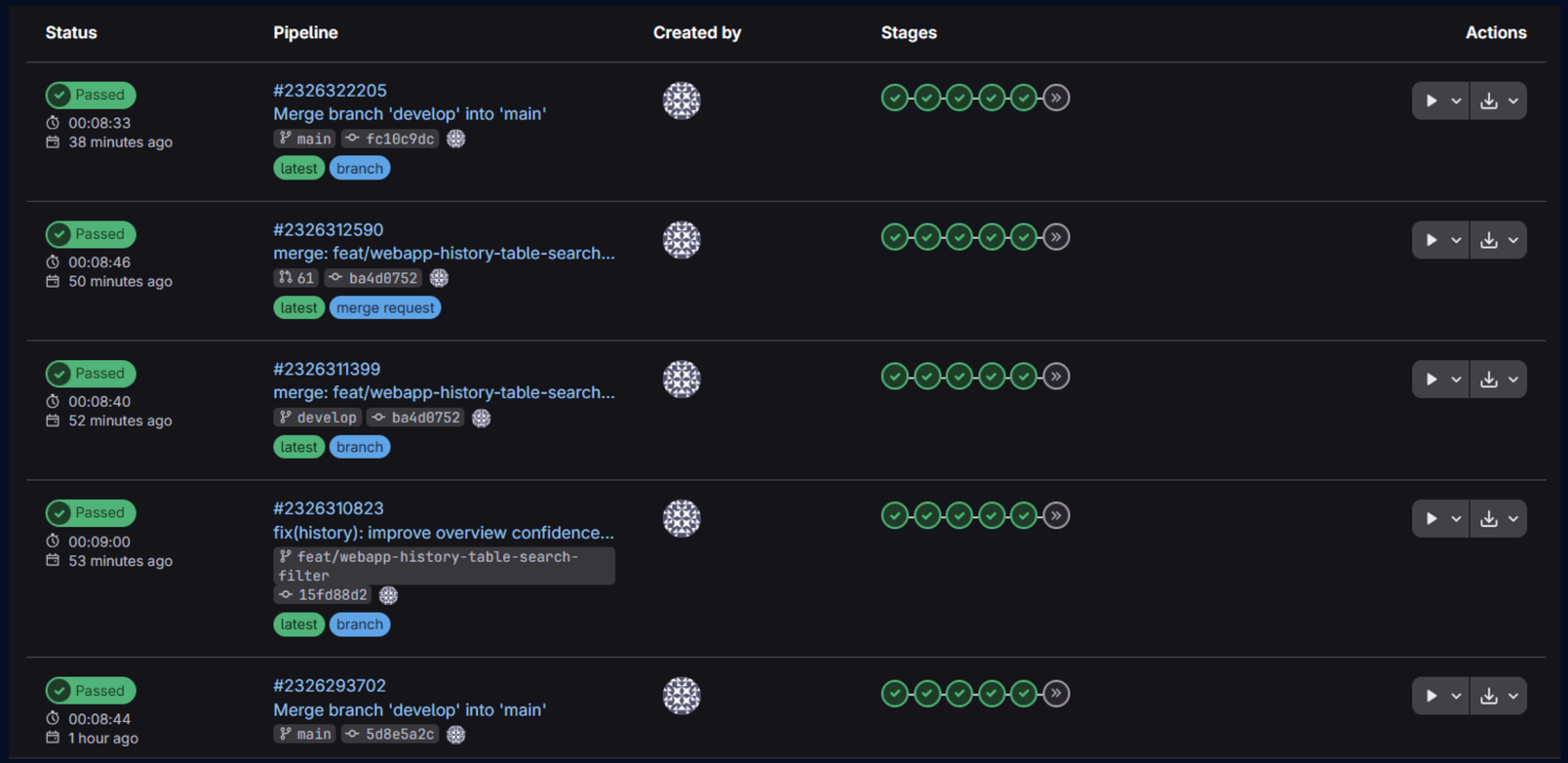
# MACHINE LEARNING OPERATIONS

## CI Pipeline Checks Before Merge




https://gitlab.com/5691-st1516-devops/ca2-daaa2b02-p2415691-gohkunming

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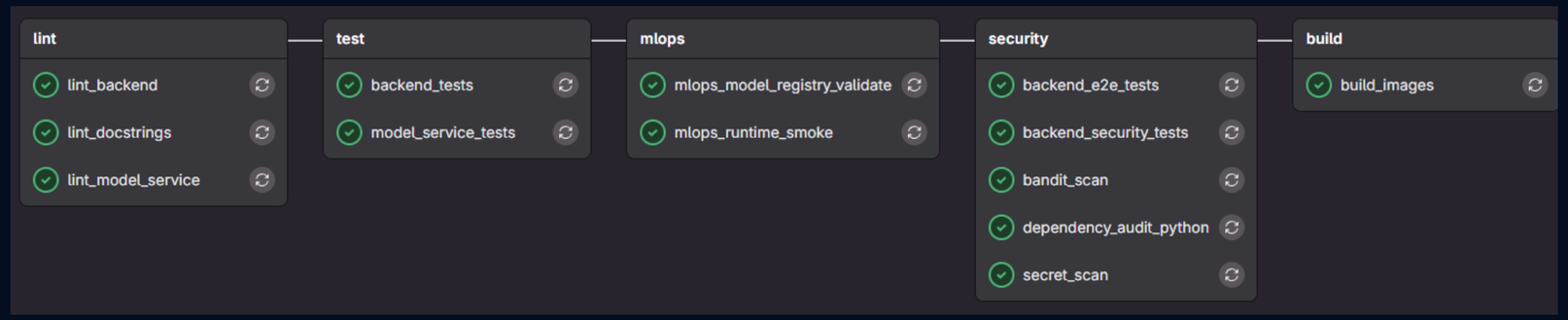
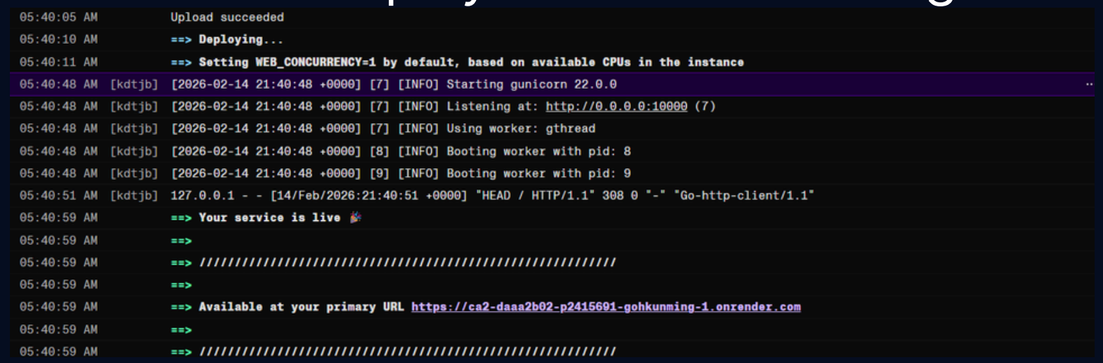
# MACHINE LEARNING OPERATIONS

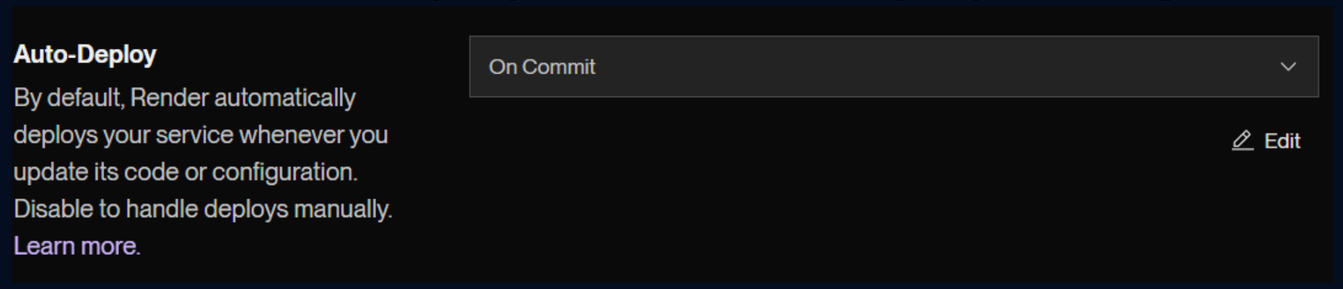
## CI Pipeline Stages




## Docker Deployment Success Logs




## Docker Deployment Auto-Deploy Config




https://gitlab.com/5691-st1516-devops/ca2-daaa2b02-p2415691-gohkunming

https://ca2-daaa2b02-p2415691-gohkunming-1.onrender.com

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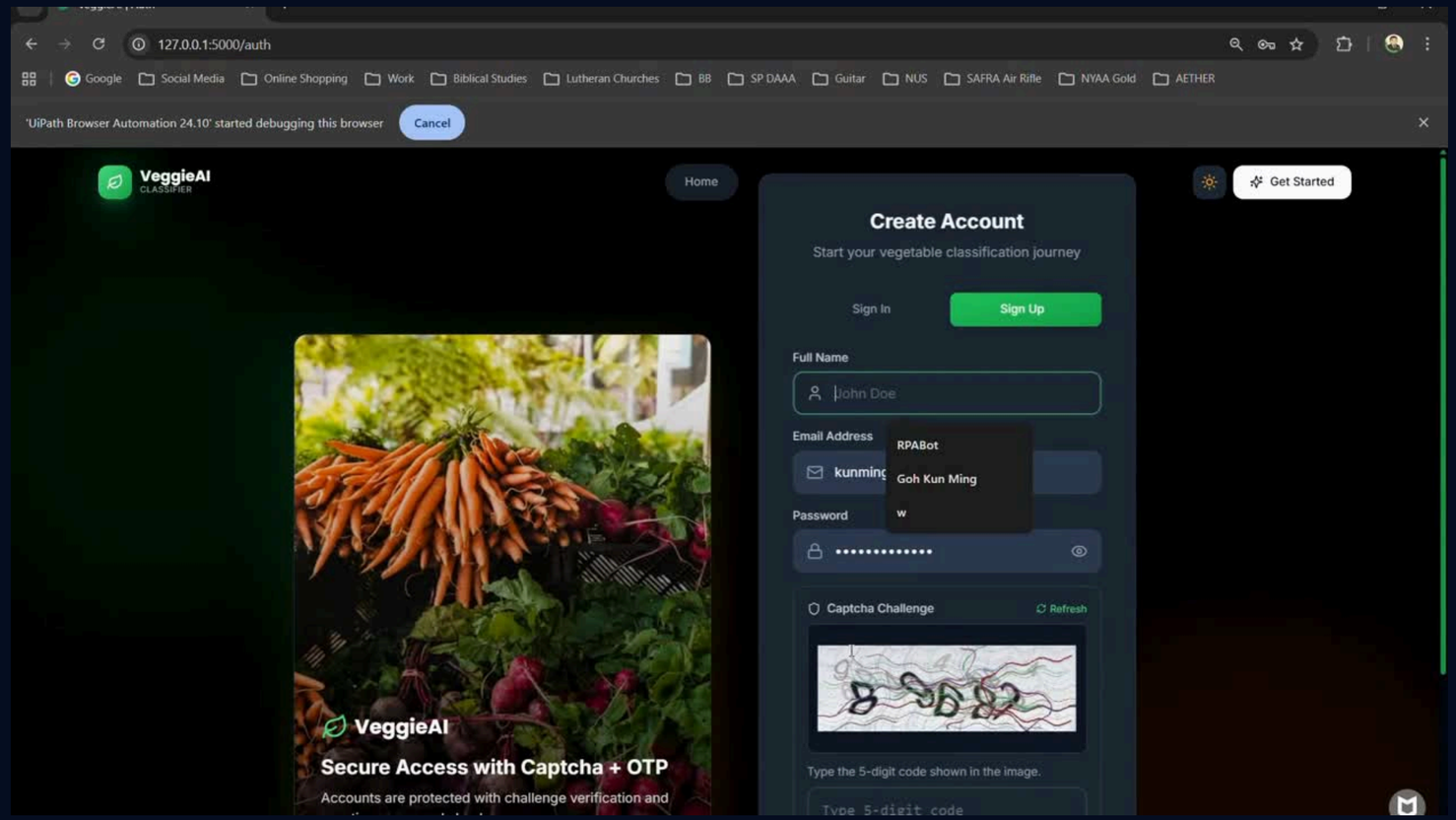
# ROBOT PROCESS AUTOMATION

## RPA For Testing of Application




RPA done on localhost so as to bypass Captcha & OTP with ‘00000’

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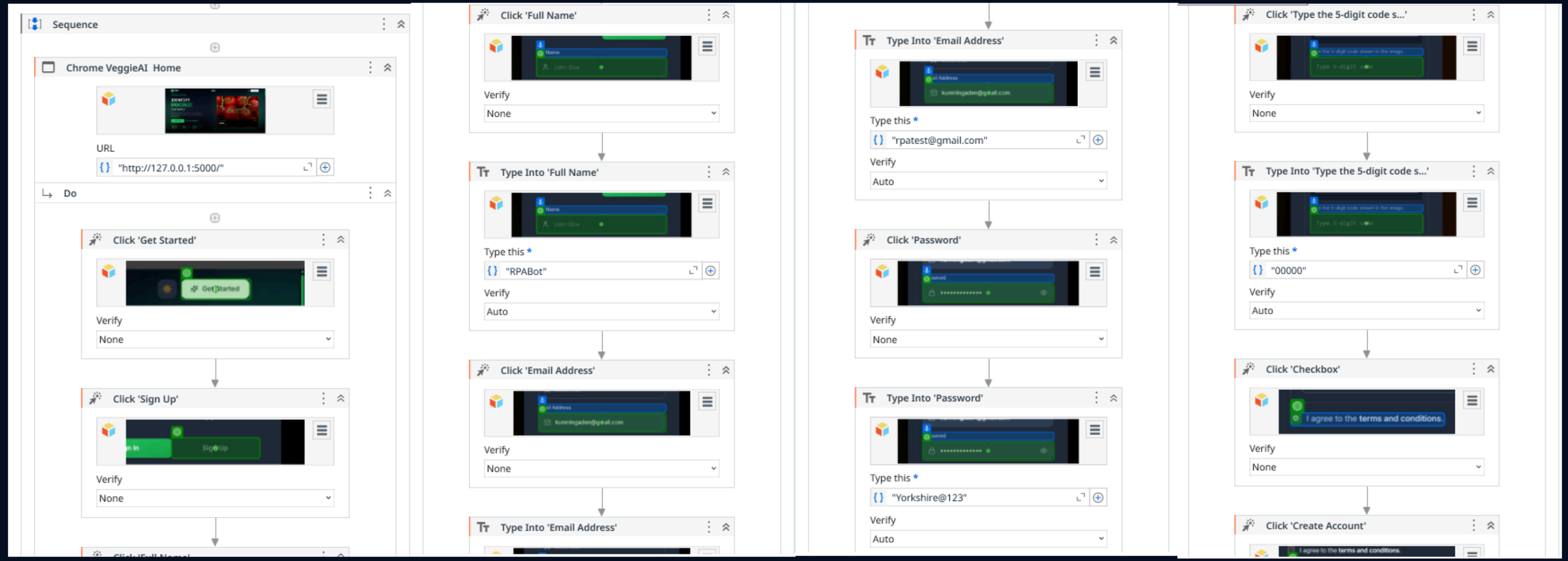
# ROBOT PROCESS AUTOMATION

## RPA Workflow



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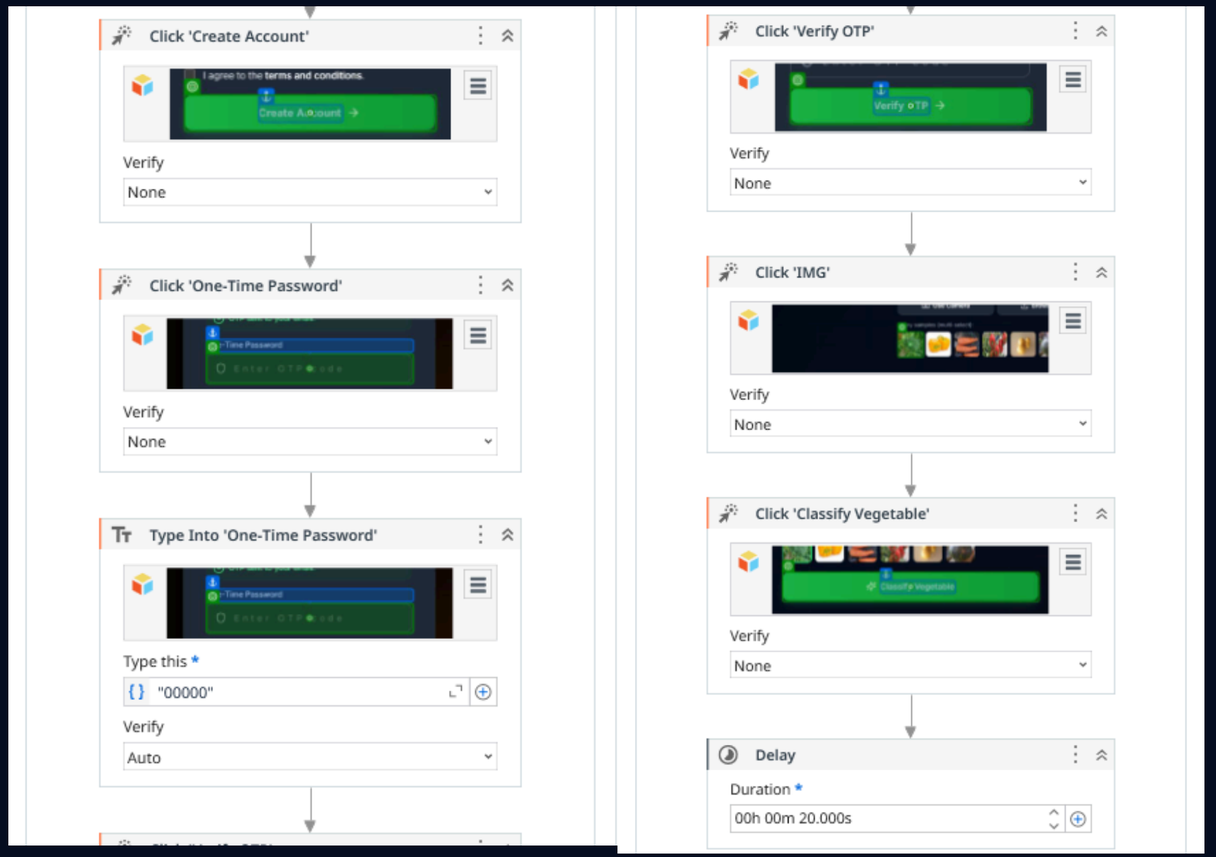
# ROBOT PROCESS AUTOMATION

## RPA Workflow



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# IMPORTANT LINKS

**GitLab**

https://gitlab.com/5691-st1516-devops/ca2-daaa2b02-p2415691-gohkunming

**Application**

https://ca2-daaa2b02-p2415691-gohkunming-1.onrender.com

**Model Serving (23x23)**

https://ca2-daaa2b02-p2415691-gohkunming.onrender.com/v1/models/vegnet-23

**Model Serving (101x101)**

https://ca2-daaa2b02-p2415691-gohkunming.onrender.com/v1/models/vegnet-101

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# REFERENCES

vegetables (2023). vegetable Object Detection Model by vegetables. [online] Roboflow. Available at: https://universe.roboflow.com/vegetables-zdlas/vegetable-cihx0 [Accessed 9 Feb. 2026].

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# THANK YOU

## FOR YOUR ATTENTION

Presented By

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Admin Number

**P2415691**